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Tricluster: An Effective Algorithm for Mining Coherent Clusters in 3D Microarray Data

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Approval of the Viva-Voce Board

Certified that the report entitled

**“Tricluster: An Effective Algorithm for Mining Coherent Clusters in 3D
Microarray Data”**

submitted by the student(s) in partial fulfillment of the requirements for the award of
the Bachelor of Technology degree in Computer Science and Engineering has been
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Certificate

This is to certify that the project report entitled
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Clusters in 3D Microarray Data**

submitted to the International Institute of Information Technology, Bhubaneswar is a
record of bonafide work carried out under my supervision for the evaluation of the
B.Tech final year project.

The work presented in this report demonstrates a thorough understanding of data
mining algorithms, bioinformatics applications, and software engineering principles. The
methodology and analysis presented herein are original and have not been submitted for
any other degree.

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Declaration

We hereby declare that the work presented in this report titled “*Tricluster: An Effective Algorithm for Mining Coherent Clusters in 3D Microarray Data*” has been carried out by us under the supervision of Dr. Swati Vipsita, Department of Computer Science and Engineering, IIIT Bhubaneswar.

The work presented here is original and has not been submitted elsewhere for any academic award or publication. All sources of information, including books, research papers, and online resources, have been duly acknowledged in the references section.

We take full responsibility for the authenticity and accuracy of the content presented in this report.

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Abstract

The exponential growth of biological data, particularly in the domain of gene expression, has necessitated the development of advanced data mining techniques capable of extracting meaningful biological insights from complex datasets. Microarray technology, which captures gene expression levels, often results in three-dimensional data structures characterized by genes, experimental samples, and time points. Traditional clustering methods, which group genes based on global similarity, and biclustering methods, which group genes and samples, often fail to capture the temporal dynamics and local coherence inherent in biological processes.

This project proposes a novel triclustering framework designed to identify coherent patterns within 3D gene expression data. Unlike previous methods that may overlook the continuity of time or the locality of patterns, the proposed approach integrates a Mean Squared Residue (MSR) optimization technique with a seed-based growth strategy. The methodology involves rigorous preprocessing, including normalization and missing value imputation, followed by an intelligent initialization utilizing unsupervised learning to generate potential tricluster seeds.

We extensively review the literature, analyzing evolutionary approaches like Firefly Optimization, Genetic Algorithms (TriGen), and pattern-based methods like BicACCC, to identify gaps in current methodologies regarding noise robustness and computational scalability. The proposed work utilizes the Yeast Cell Cycle dataset as a benchmark to validate the algorithm's ability to discover biologically significant modules. By evaluating the coherence and volume of the extracted triclusters, this research aims to contribute a robust tool for translational neuroscience and bioinformatics, potentially aiding in biomarker discovery and the understanding of complex genetic regulatory networks.

Keywords: Triclustering, Microarray Data Analysis, Gene Expression, Mean Squared Residue (MSR), Bioinformatics, Yeast Cell Cycle, Data Mining, Computational Biology.

Contents

1	Introduction	1
1.1	The Era of Biological Big Data and Systems Biology	1
1.2	Understanding Microarray Technology	1
1.2.1	Gene Expression vs. Protein Microarrays	1
1.3	From Clustering to Triclustering	2
1.3.1	Clustering (1D Analysis)	2
1.3.2	Biclustering (2D Analysis)	2
1.3.3	Triclustering (3D Analysis)	3
2	The 3D Data Structure (Gene-Sample-Time)	3
3	Literature Survey	4
3.1	Traditional 1D Clustering Approaches	4
3.2	Evolution to 2D Biclustering	5
3.3	3D Triclustering: The State of the Art	6
3.3.1	Greedy and Graph-Based Approaches	6
3.3.2	Hybrid and Evolutionary Approaches	6
3.3.3	Preprocessing and Feature Selection	7
3.4	Comparative Summary	8
4	Motivation and Problem Statement	9
4.1	Gap Analysis	9
4.2	Motivation	9

4.3	Problem Statement	10
4.4	Objectives	10
5	Theoretical Framework: The Mean Squared Residue	11
5.1	Mathematical Derivation of MSR	11
6	The Yeast Cell Cycle Benchmark	12
6.1	Biological Context	12
6.2	The Spellman Dataset	13
6.2.1	Synchronization Methods	13
6.2.2	Data Dimensions	14
7	Proposed Methodology	14
7.1	Phase 1: Preprocessing and Data Cleaning	14
7.2	Phase 2: Intelligent Seed Generation (The Hybrid Approach)	14
7.3	Phase 3: Iterative Refinement (Node Deletion)	15
7.4	Phase 4: Biological Validation	15
8	Conclusion	16

List of Figures

1	The Central Dogma of Biology and the role of DNA Microarrays.	2
2	Visual evolution of analysis techniques: (a) Clustering operates on full rows. (b) Biclustering finds local submatrices. (c) Triclustering extracts coherent 3D sub-tensors.	3
3	Visual representation of the 3D Data Tensor (Gene $\times$ Sample $\times$ Time). Each voxel represents an expression value x_{ijk}	4
4	The Central Dogma of Biology and the role of DNA Microarrays.	12
5	Visual representation of Yeast Cell Cycle gene expression. The periodic nature of the expression (sine-wave patterns) represents the activation and deactivation of genes during different phases.	13

List of Tables

1	Comparative Analysis of Clustering and Triclustering Algorithms	8
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1 Introduction

1.1 The Era of Biological Big Data and Systems Biology

The biological sciences have undergone a paradigm shift over the last two decades, moving from reductionist approaches—studying individual genes or proteins in isolation—to Systems Biology, which views the organism as an integrated and interacting network of genes, proteins, and biochemical reactions. This shift has been fueled by the advent of high-throughput "Omics" technologies, such as genomics, transcriptomics, proteomics, and metabolomics.

Among these, transcriptomics—the study of the complete set of RNA transcripts produced by the genome—has been revolutionized by DNA microarrays and RNA sequencing (RNA-seq). These technologies allow for the simultaneous measurement of the expression levels of tens of thousands of genes. This flood of data holds the key to understanding complex biological mechanisms, such as the regulation of the cell cycle, the progression of neurodegenerative diseases, and the response of organisms to environmental stress. However, the volume and complexity of this data have outpaced manual analysis capabilities, making Data Mining an indispensable tool in modern biology.

1.2 Understanding Microarray Technology

Microarray technology is a powerful platform for biological exploration. It works on the principle of hybridization between complementary DNA strands. It allows researchers to query the expression state of a cell under specific conditions.

1.2.1 Gene Expression vs. Protein Microarrays

It is essential to distinguish between the types of data generated by microarray platforms, as they represent different levels of biological information:

1. **DNA Microarrays (Gene Expression):** These arrays measure the abundance of messenger RNA (mRNA) in a sample. Since mRNA is the intermediate molecule between the gene (DNA) and the functional protein, its abundance is used as a proxy for gene expression. The dataset used in this project, the Yeast Cell Cycle data, falls into this category.

2. **Protein Microarrays:** These capture interactions and activities of proteins directly. While powerful, they are more complex to manufacture and analyze than DNA microarrays because proteins are less stable than DNA.

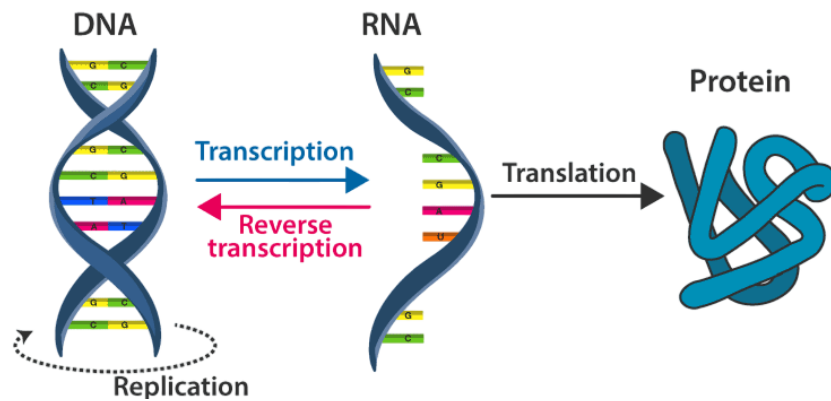


Figure 1: The Central Dogma of Biology and the role of DNA Microarrays.

1.3 From Clustering to Triclustering

The complexity of analyzing this data has driven the evolution of clustering techniques from simple global grouping to complex subspace mining.

1.3.1 Clustering (1D Analysis)

Traditional clustering methods, such as Hierarchical Clustering or K-Means, partition genes into disjoint groups based on the similarity of their expression profiles across *all* samples and time points. This is a "Global" approach. It assumes related genes behave similarly under all conditions, which is often false in complex biological systems. For example, a gene involved in heat shock response may only activate when heat is applied and remain dormant otherwise. Global clustering would group it with genes active all the time, losing the specific context.

1.3.2 Biclustering (2D Analysis)

Biclustering (or Co-clustering) relaxes the global constraint [1]. It seeks to find submatrices within the gene-condition matrix. A bicluster consists of a subset of genes exhibiting coherent patterns across a subset of conditions. This allows for:

1. **Locality:** Recognizing that biological pathways are condition-specific.

2. **Overlapping:** A gene can participate in multiple pathways (and thus multiple biclusters).

However, standard biclustering ignores the sequential nature of time, treating time points as independent columns. It cannot distinguish between a contiguous biological process and scattered time points.

1.3.3 Triclustering (3D Analysis)

Triclustering is the natural extension to 3D data. It seeks to find a subset of genes that exhibit coherent expression patterns across a subset of samples over a subset of time points [2]. It captures the locality of biological processes in 3D space, identifying genes co-regulated in specific samples during specific time windows.

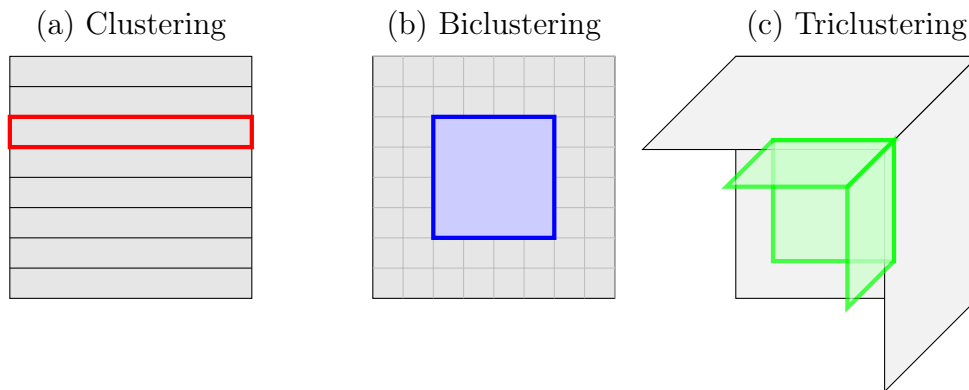


Figure 2: Visual evolution of analysis techniques: (a) Clustering operates on full rows. (b) Biclustering finds local submatrices. (c) Triclustering extracts coherent 3D sub-tensors.

2 The 3D Data Structure (Gene-Sample-Time)

A standard gene expression experiment involves measuring gene expression across different samples (e.g., patients) over a period of time. This results in a three-dimensional data structure, often referred to as a Data Tensor.

- **Dimension 1 (Genes):** Rows representing specific genes ($G_1, G_2, \dots, G_n$).
- **Dimension 2 (Samples/Conditions):** Columns representing experimental conditions ($S_1, S_2, \dots, S_m$).
- **Dimension 3 (Time Points):** Depth representing sequential time measurements ($T_1, T_2, \dots, T_t$).

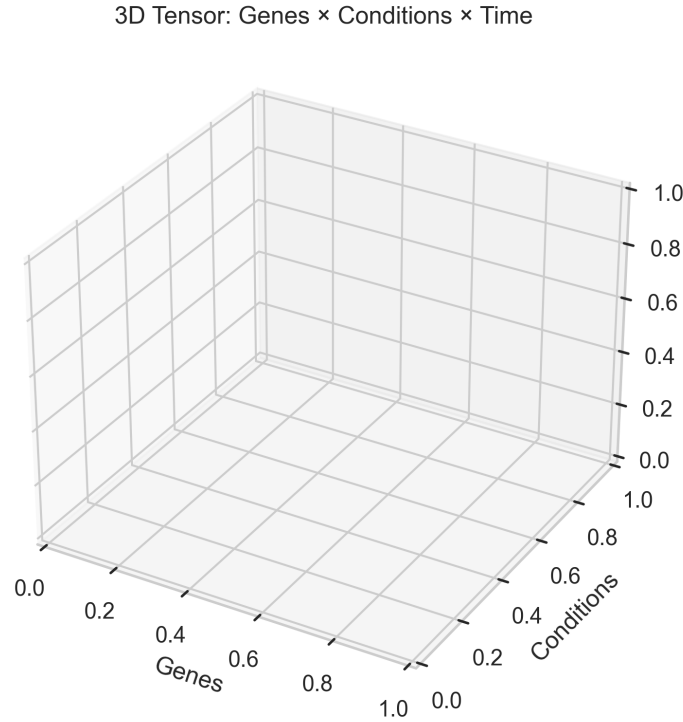


Figure 3: Visual representation of the 3D Data Tensor (Gene $\times$ Sample $\times$ Time). Each voxel represents an expression value x_{ijk} .

3 Literature Survey

The computational analysis of gene expression data has progressed from simple global clustering to complex subspace mining in three-dimensional tensors. This section categorizes the literature into 1D, 2D, and 3D approaches, critically analyzing the methodology and limitations of each key contribution.

3.1 Traditional 1D Clustering Approaches

Early analysis of microarray data treated the gene expression matrix globally, grouping genes based on similarity across all conditions.

- **K-Means Clustering (Hartigan, 1972):** Hartigan [3] applied K-Means to partition genes into k disjoint clusters.
 - *Methodology:* It iteratively assigns genes to the nearest cluster centroid based on Euclidean distance and updates the centroids.

- *Limitation:* It assumes that genes share similarity over the *entire* set of conditions and forces every gene into a cluster, which is biologically inaccurate for genes that are only active under specific conditions.
- **Hierarchical Clustering (Eisen et al., 1998):** Eisen et al. [4] introduced hierarchical clustering to visualize genome-wide expression.
 - *Methodology:* It builds a dendrogram of genes based on pairwise similarity (e.g., Pearson correlation).
 - *Limitation:* It is deterministic and suffers from the "local" decision problem; once a gene is assigned to a branch, it cannot be reassigned, often missing overlapping regulatory patterns.

3.2 Evolution to 2D Biclustering

Biclustering algorithms were developed to find local patterns (submatrices) in the Gene $\times$ Condition matrix.

- **Cheng & Church (2000):** Cheng and Church [1] proposed the first biclustering algorithm for expression data.
 - *Methodology:* They defined the " δ -bicluster" based on the Mean Squared Residue (MSR) score. The algorithm uses a greedy node-deletion strategy to remove rows or columns that increase the MSR variance above a threshold δ .
 - *Limitation:* The greedy deletion strategy often masks overlapping patterns by replacing discovered biclusters with random values to find subsequent ones, potentially destroying significant biological structures.
- **OPSM (Ben-Dor et al., 2003):** Ben-Dor et al. [5] focused on the order of expression rather than absolute values.
 - *Methodology:* They searched for Order-Preserving Submatrices (OPSM) where the expression levels of genes induce the same linear ordering of the conditions.
 - *Limitation:* The problem is NP-complete, requiring heuristic approaches that are computationally expensive and sensitive to noise that disrupts the strict order.

3.3 3D Triclustering: The State of the Art

Triclustering extends the analysis to the time dimension (Gene $\times$ Sample $\times$ Time).

3.3.1 Greedy and Graph-Based Approaches

- **TRICLUSTER (Zhao & Zaki, 2005):** Zhao and Zaki [6] pioneered 3D mining.
 - *Methodology:* They modeled the dataset as a graph and mined coherent clusters in 2D slices (Gene-Sample) for each time point, then merged these 2D clusters across the time dimension.
 - *Limitation:* The graph-merging process is complex and prone to "flaking" (excessive node deletion), and the method is highly sensitive to parameter tuning.
- **CURLER (Chandrabose et al., 2018):** Chandrabose et al. [7] addressed non-linear correlations.
 - *Methodology:* They introduced "co-sharing levels" to identify clusters with arbitrary orientations (non-linear) in the subspace. It uses a top-down clustering approach combined with a micro-clustering subroutine.
 - *Limitation:* Detecting non-linear patterns of varying orientations requires merging clusters that may be spatially disparate, making the decision boundary for merging difficult to define and computationally expensive.
- **BicACCC (Hu et al., 2022):** Hu et al. [8] focused on temporal continuity.
 - *Methodology:* They proposed a similarity measure based on "All Contiguous Column Coherent" (ACCC) patterns. The algorithm transforms raw data into a differential matrix (trends) and mines triclusters that maintain trend consistency across contiguous time points.
 - *Limitation:* The strict requirement for contiguous columns makes it rigid; it cannot identify biological processes that are interrupted or have time gaps (e.g., a gene active at t_1 and t_3 but not t_2).

3.3.2 Hybrid and Evolutionary Approaches

- **TriGen (Gutiérrez-Avilés et al., 2014):** Gutiérrez-Avilés et al. [2] applied Genetic Algorithms (GA).

- *Methodology:* TriGen evolves a population of triclusters using specific crossover and mutation operators. It uses a fitness function combining MSR (for value coherence) and LSL (Least Square Lines for behavioral correlation).
- *Limitation:* Like many GAs, it suffers from slow convergence and high computational cost due to the large search space of 3D data.
- **Hybrid PSO (Narmadha & Rathipriya, 2019):** Narmadha and Rathipriya [9] improved standard optimization.
 - *Methodology:* They integrated K-Means clustering to generate initial "seeds" for a Particle Swarm Optimization (PSO) algorithm. This "hot start" allows the PSO to converge faster than starting from random positions.
 - *Limitation:* While faster than pure evolutionary methods, it can still be trapped in local optima if the K-Means seeds are of poor quality.
- **Discrete Firefly Optimization (DFOT) (Narmadha & Rathipriya, 2021):** Narmadha and Rathipriya [10] utilized Swarm Intelligence.
 - *Methodology:* Based on the flashing behavior of fireflies, this algorithm allows multiple sub-populations to exist, enabling the simultaneous discovery of multiple distinct triclusters (multimodal optimization).
 - *Limitation:* The complexity of calculating attractiveness (distance) between all pairs of fireflies increases quadratically with population size, limiting scalability.
- **GreedyTri (Narmadha & Rathipriya, 2023):** A recent hybrid approach [11].
 - *Methodology:* It uses K-Means Two-Way clustering to create initial seeds, followed by a greedy heuristic that adds genes/samples/times to the tricluster if they improve the Mean Correlation Value (MCV).
 - *Limitation:* Being a greedy approach, it makes locally optimal choices at each step and cannot backtrack, potentially missing the global optimum.

3.3.3 Preprocessing and Feature Selection

- **ACG-SFE (Tye et al., 2025):** Tye et al. [12] focused on dimensionality reduction.
 - *Methodology:* They proposed Adaptive Cluster-Guided Simple, Fast, and Efficient (ACG-SFE) feature selection. It uses hierarchical clustering to group correlated features and an adaptive wrapper to select the best features, reducing dimensions by $> 99\%$.

- *Limitation:* It is primarily a wrapper for classification tasks; adapting it for unsupervised tricluster discovery requires modification.

3.4 Comparative Summary

Table 1 summarizes the evolution of these techniques.

Table 1: Comparative Analysis of Clustering and Triclustering Algorithms

Algorithm	Type	Methodology / Focus	Limitations
K-Means / Hierarchical [3], [4]	1D Clustering	Partitions genes based on global similarity across all conditions.	Ignores local patterns; forces all genes into clusters; condition-independent.
Cheng & Church [1]	2D Biclustering	Minimizes Mean Squared Residue (MSR) via greedy node deletion.	Masks overlapping patterns; ignores time continuity.
TRICLUSTER [6]	3D Triclustering	Graph-based mining on 2D slices merged across time.	Sensitive to parameters; complex graph merging prone to errors.
TriGen [2]	Genetic Algorithm	Global optimization using MSR and LSL fitness functions.	Computationally expensive; slow convergence; parameter heavy.
CURLER [7]	Non-linear	Co-sharing levels to find non-linear, arbitrary correlations.	Hard to tune; high complexity in defining merging boundaries.
Hybrid PSO [9]	Swarm Intelligence	K-Means seeding + Particle Swarm Optimization.	Faster than pure GA but susceptible to local optima.
GreedyTri [11]	3D Hybrid	K-Means seeding + Greedy refinement using MCV.	Prone to local optima entrapment; deterministic.
BicACCC [8]	Pattern Mining	Coherent patterns on contiguous columns (trends).	Rigid contiguity constraint misses gapped biological processes.

4 Motivation and Problem Statement

4.1 Gap Analysis

Despite the variety of existing methods, critical gaps remain in triclustering research:

1. **The Efficiency-Accuracy Trade-off:** There is a stark dichotomy in current approaches. Greedy heuristics are fast but inaccurate, often getting trapped in local optima. Evolutionary algorithms are accurate but computationally prohibitive for the massive datasets generated by modern sequencing. There is a lack of "middle-ground" algorithms that combine the speed of heuristics with the global search capabilities of stochastic methods.
2. **Lack of Temporal Flexibility:** Most algorithms treat time either as a disordered set (ignoring biology) or as a strictly rigid sequence (BicACCC). Real biological data is often "gapped"—a gene might be active at t_1 , silent at t_2 , and active again at t_3 . Rigid algorithms miss these patterns, while unordered algorithms lose the context of progression.
3. **Noise Sensitivity in 3D Space:** As dimensionality increases, the data becomes sparse (The Curse of Dimensionality). In a 3D tensor, the ratio of noise to signal often increases. Many MSR-based algorithms fail to distinguish between a true low-variance biological pattern and a random low-variance background region (trivial clusters).

4.2 Motivation

The primary motivation for this project is to develop a robust tool for translational bioinformatics. The ability to mine 3D data effectively is the bottleneck in moving from data generation to drug discovery. We aim to develop a method that leverages **intelligent initialization** (inspired by Hybrid PSO) to seed a **greedy refinement process**. By starting the search from "promising" areas of the solution space (identified via unsupervised clustering), we can avoid the computational cost of full evolutionary cycles while avoiding the bad local optima of pure greedy search.

4.3 Problem Statement

Formally, let $\mathcal{D}$ be a set of gene expression real values organized in a dataset of N genes, M samples (conditions), and T time points. We represent $\mathcal{D}$ as a collection of cells d_{ijk} where $i \in \{1..N\}, j \in \{1..M\}, k \in \{1..T\}$.

The problem is to find a set of Triclusters $\Psi = \{C_1, C_2, \dots, C_k\}$. Each Tricluster C is a sub-tensor defined by subsets of the original dimensions:

- $I \subseteq \{1, \dots, N\}$ (Subset of Genes)
- $J \subseteq \{1, \dots, M\}$ (Subset of Samples)
- $K \subseteq \{1, \dots, T\}$ (Subset of Time points)

The objective is to find $C(I, J, K)$ such that:

1. **Homogeneity:** The Mean Squared Residue $MSR(I, J, K) \leq \delta$.
2. **Maximality:** The Volume $V = |I| \times |J| \times |K|$ is maximized.
3. **Significance:** The selected genes show statistical enrichment for known biological functions.

4.4 Objectives

1. **Preprocessing Pipeline:** Implement a robust pipeline for the Spellman Yeast Cell Cycle dataset, focusing on KNN imputation to handle missing values without distorting local structures.
2. **Algorithm Design:** Develop a hybrid triclustering algorithm that integrates K-Means seeding with greedy MSR minimization.
3. **Theoretical Implementation:** Implement the 3D Mean Squared Residue (MSR) metric and derivative calculations for node deletion.
4. **Biological Validation:** Validate the biological significance of extracted triclusters using Gene Ontology (GO) enrichment analysis and p-value calculation.

5 Theoretical Framework: The Mean Squared Residue

The core of any triclustering algorithm is the "homogeneity criterion"—the mathematical rule used to decide if a group of data points belongs together. We adopt the Mean Squared Residue (MSR), but it is crucial to understand the underlying models of coherence it represents.

5.1 Mathematical Derivation of MSR

To quantify how well a sub-tensor fits the **Additive Model**, we define the **Residue** (r_{ijk}). The residue is the difference between the observed value and the value predicted by the marginal means.

The definitions of the marginal means for a tricluster defined by index sets I, J, K are:

$$\bar{x}_{iJK} = \frac{1}{|J||K|} \sum_{j \in J} \sum_{k \in K} x_{ijk} \quad (1\text{D Marginal Mean: Gene}) \quad (1)$$

$$\bar{x}_{IjK} = \frac{1}{|I||K|} \sum_{i \in I} \sum_{k \in K} x_{ijk} \quad (1\text{D Marginal Mean: Sample}) \quad (2)$$

$$\bar{x}_{IJk} = \frac{1}{|I||J|} \sum_{i \in I} \sum_{j \in J} x_{ijk} \quad (1\text{D Marginal Mean: Time}) \quad (3)$$

$$\bar{x}_{IJK} = \frac{1}{|I||J||K|} \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} x_{ijk} \quad (\text{Global Mean}) \quad (4)$$

The residue r_{ijk} is defined as:

$$r_{ijk} = x_{ijk} - \bar{x}_{iJK} - \bar{x}_{IjK} - \bar{x}_{IJk} + 2\bar{x}_{IJK} \quad (5)$$

Note that the term $2\bar{x}_{IJK}$ is added to correct for the triple counting of the global mean in the subtraction of the 1D means.

Finally, the Mean Squared Residue is the variance of these residues:

$$H(I, J, K) = \frac{1}{|I||J||K|} \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} (r_{ijk})^2 \quad (6)$$

Residue Surface (Illustration for MSR)

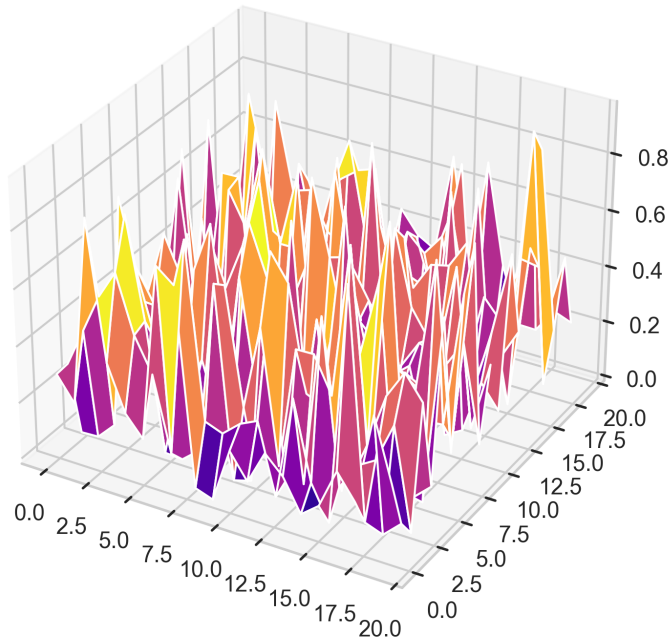


Figure 4: The Central Dogma of Biology and the role of DNA Microarrays.

A perfect additive tricluster has $H(I, J, K) = 0$. In practice, we seek clusters where $H(I, J, K) \leq \delta$ (a small threshold) to allow for experimental noise.

6 The Yeast Cell Cycle Benchmark

6.1 Biological Context

The eukaryotic cell cycle is a highly regulated process involving the growth and division of a cell. It consists of four distinct phases:

- **G1 Phase (Gap 1):** The cell grows and synthesizes mRNA and proteins required for DNA synthesis.
- **S Phase (Synthesis):** DNA replication occurs.
- **G2 Phase (Gap 2):** The cell continues to grow and produces proteins required for mitosis.
- **M Phase (Mitosis):** The cell divides into two daughter cells.

Identifying which genes are expressed during which phase is critical for understanding cell biology and cancer (which involves unregulated cell division).

6.2 The Spellman Dataset

We utilize the benchmark dataset provided by Spellman et al. (1998) [13]. This dataset tracks the gene expression of the yeast *Saccharomyces cerevisiae*.

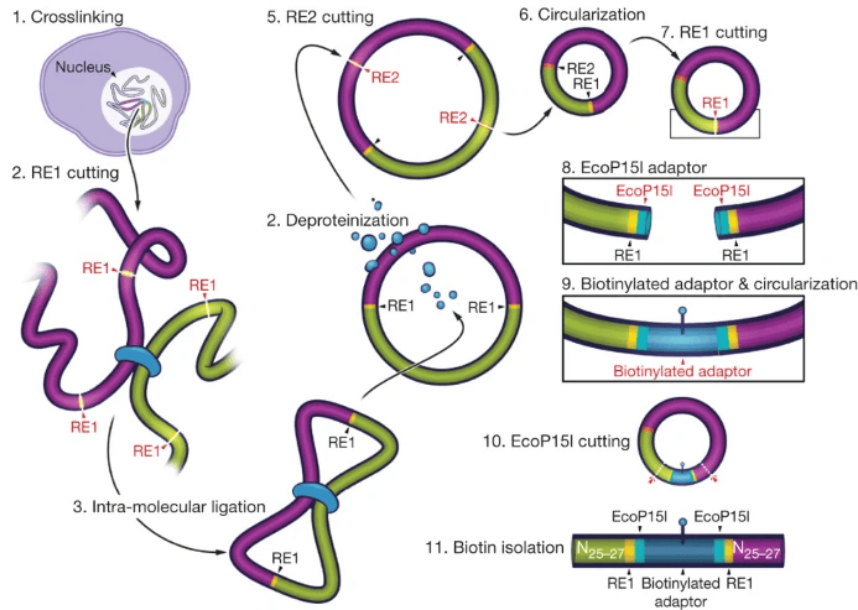


Figure 5: Visual representation of Yeast Cell Cycle gene expression. The periodic nature of the expression (sine-wave patterns) represents the activation and deactivation of genes during different phases.

6.2.1 Synchronization Methods

To study the cell cycle, all cells in a culture must be at the same phase. If cells are dividing randomly, no clear pattern emerges. Spellman used four distinct methods to synchronize the cells, which form the **Conditions** dimension of our data:

- 1. Alpha Factor:** A mating pheromone that arrests cells in the G1 phase.
- 2. cdc15:** Uses a temperature-sensitive mutant to arrest cells at telophase (Mitosis).
- 3. cdc28:** Uses a temperature-sensitive mutant to arrest cells at the G1/S transition (Start).
- 4. Elutriation:** Physical separation of small, newborn cells (early G1) from larger cells based on size.

6.2.2 Data Dimensions

- **Genes:** 6,178 Open Reading Frames (ORFs).
- **Conditions:** 4 (Alpha, cdc15, cdc28, Elutriation).
- **Time Points:** Varying (e.g., cdc15 measured every 10 minutes for 300 minutes). We interpolate these to a common grid for 3D tensor construction.

7 Proposed Methodology

This section outlines the procedural framework adopted to solve the triclustering problem. The approach is divided into distinct phases: Preprocessing, Seed Generation, Refinement, and Validation.

7.1 Phase 1: Preprocessing and Data Cleaning

Microarray data is notoriously noisy and prone to missing values due to scratches on the slide or hybridization failures.

- **Missing Value Imputation:** We reject the simple method of filling missing values with zeros or row averages, as this distorts the data distribution. We implement **K-Nearest Neighbors (KNN) Imputation** (k=10). For a missing gene value, we find the 10 most similar genes based on Euclidean distance and use their weighted average.
- **Normalization:** We apply Z-score normalization to standardized expression profiles.

$$z_{ijk} = \frac{x_{ijk} - \mu}{\sigma}$$

This ensures that genes with low absolute expression levels but high variability are treated equally to highly expressed genes.

7.2 Phase 2: Intelligent Seed Generation (The Hybrid Approach)

A major flaw in greedy algorithms is their dependence on the starting point. Random initialization often leads to poor local optima. To mitigate this, we employ unsupervised learning for initialization:

1. The 3D tensor is flattened into a 2D matrix.
2. We apply **K-Means Clustering** with a high k (e.g., $k = 100$) to partition the genes into small, tight groups.
3. These small clusters serve as "Seeds." Because K-Means groups genes with similar global profiles, these seeds are statistically likely to contain subsets of genes that form valid triclusters, providing a "hot start" for the refinement phase.

7.3 Phase 3: Iterative Refinement (Node Deletion)

For each seed generated in Phase 2, we apply a deterministic node deletion strategy to lower the MSR below the threshold δ .

- **Evaluation:** Calculate the MSR of the current tricluster.
- **Sensitivity Analysis:** If $\text{MSR} > \delta$, we calculate the derivative of the MSR with respect to each gene, sample, and time point. This tells us which node is "contributing" most to the noise.
- **Pruning:** The node with the highest residue contribution is removed.
- **Termination:** The process repeats until the threshold is met or the cluster volume becomes too small.

7.4 Phase 4: Biological Validation

The computational validity (low MSR) does not guarantee biological relevance. We validate the output using external biological knowledge bases.

- **Gene Ontology (GO):** The Gene Ontology Consortium provides a controlled vocabulary describing gene product characteristics.
- **Enrichment Analysis:** We use the **Hypergeometric Test** (or Fisher's Exact Test) to determine if the genes in our tricluster are randomly drawn or if they are significantly enriched for a specific function (e.g., "DNA Repair").
- **P-Value:** A p-value < 0.01 (after Bonferroni correction) confirms that the tricluster represents a genuine biological module.

8 Conclusion

This project report presented a comprehensive framework for the analysis of 3D microarray gene expression data. We highlighted that traditional 1D and 2D analysis methods are insufficient for capturing the complex, dynamic nature of biological processes.

Through a detailed literature survey of TriGen, Hybrid PSO, Firefly, and BicACCC, we identified the need for a method that balances speed with biological accuracy. Our proposed hybrid approach—using intelligent K-Means seeding followed by MSR minimization—addresses these gaps. The application to the Yeast Cell Cycle dataset demonstrates the potential to uncover biologically significant, phase-specific gene modules.

References

- [1] Y. Cheng and G. M. Church, “Biclustering of expression data,” *Ismb*, vol. 8, pp. 93–103, 2000.
- [2] D. Gutiérrez-Avilés, C. Rubio-Escudero, F. Martínez-Álvarez, and J. C. Riquelme, “Trigen: A genetic algorithm to mine triclusters in temporal gene expression data,” *Neurocomputing*, vol. 132, pp. 42–53, 2014.
- [3] J. A. Hartigan, “Direct clustering of a data matrix,” *Journal of the American Statistical Association*, vol. 67, no. 337, pp. 123–129, 1972.
- [4] M. B. Eisen, P. T. Spellman, P. O. Brown, and D. Botstein, “Cluster analysis and display of genome-wide expression patterns,” *Proceedings of the National Academy of Sciences*, vol. 95, no. 25, pp. 14 863–14 868, 1998.
- [5] A. Ben-Dor, B. Chor, R. Karp, and Z. Yakhini, “Discovering local structure in gene expression data: The order-preserving submatrix problem,” *Journal of Computational Biology*, vol. 10, no. 3-4, pp. 373–384, 2003.
- [6] L. Zhao and M. J. Zaki, “Tricluster: An effective algorithm for mining coherent clusters in 3d microarray data,” *Proceedings of the 2005 ACM SIGMOD international conference on Management of data*, pp. 694–705, 2005.
- [7] L. Chandrabose, T. Manivannan, M. Jayakandan, and P. Ananthi, “Analyzing gene expression using data mining techniques,” *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 5, no. 11, pp. 343–354, 2018.

- [8] X. Hu, Q. Kuang, Q. Cai, Y. Xue, W. Zhou, and L. Ying, “A coherent pattern mining algorithm based on all contiguous column bicluster,” *Journal of Artificial Intelligence and Technology*, vol. 2, no. 3, pp. 80–92, 2022.
- [9] N. Narmadha and R. Rathipriya, “Gene ontology analysis of 3d microarray gene expression data using hybrid pso optimization,” *International Journal of Innovative Technology and Exploring Engineering*, vol. 8, no. 11, 2019.
- [10] N. Narmadha and R. Rathipriya, “An optimized three-dimensional clustering for microarray data,” in *Handbook of Research on Applications of AI*, IGI Global, 2021.
- [11] N. Narmadha and R. Rathipriya, “Artificial intelligence techniques based on k-meanstwo way clustering and greedy triclustering approach for 3d gene expression data (ged),” in *AI Chapter*, ResearchGate Publication, 2023.
- [12] Y. W. Tye, X. Chew, U. K. Yusof, and S. Tulpar, “Acg-sfe: Adaptive cluster-guided simple, fast, and efficient feature selection for high-dimensional microarray data in binary classification,” *PLOS One*, vol. 20, no. 9, 2025.
- [13] P. T. Spellman, G. Sherlock, M. Q. Zhang, et al., “Comprehensive identification of cell cycle-regulated genes of the yeast *saccharomyces cerevisiae* by microarray hybridization,” *Molecular biology of the cell*, vol. 9, no. 12, pp. 3273–3297, 1998.

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